

Data Communications and Networking

Chapter 1

Introduction

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1-1 DATA COMMUNICATIONS

Telecommunication-communication at a distance.

Data -information presented in whatever form is agreed upon by the parties creating and using the data.

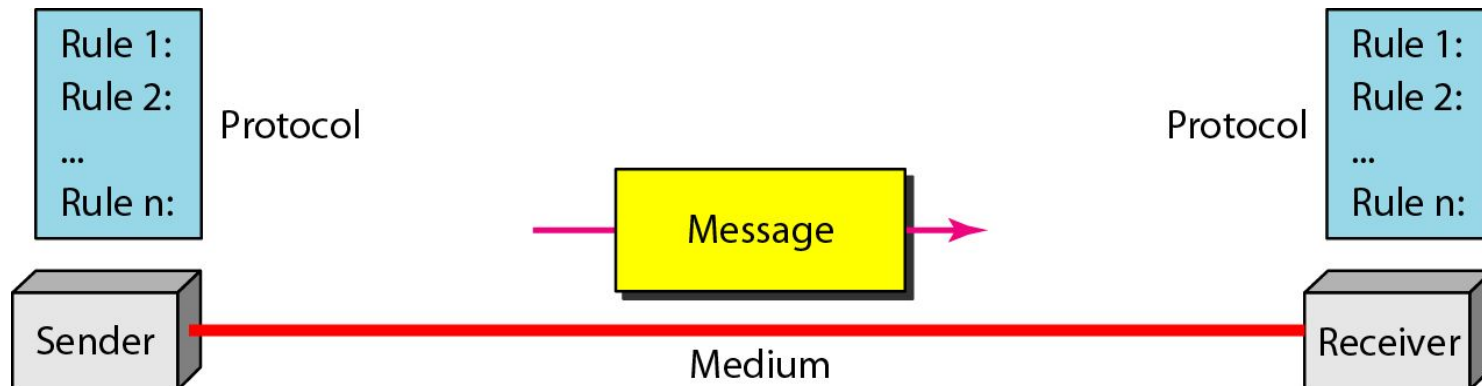
Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Fundamental characteristics[Effectiveness of DC]:

1. **Delivery**-correct destination
2. **Accuracy**-deliver data accurately
3. **Timeliness**- delivery data in timely manner
4. **Jitter**-variation in packet arrival time.

Components of a data communication system

- **Message**- information to be communicated
- **Sender**- send the data message
- **Receiver** – receives the message
- **Transmission Medium**- physical path by which message travels from sender to receiver.
- **Protocol**- set of rules govern the data communications



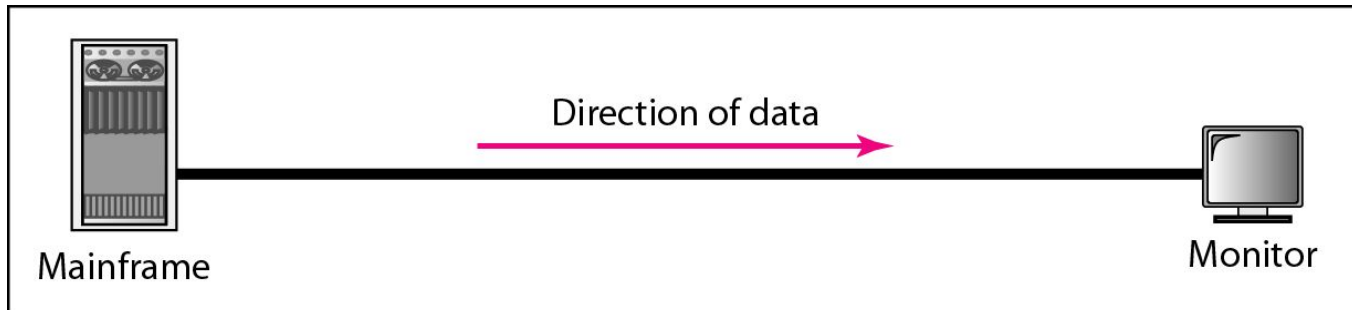
Data Representation

- **Text**
 - Represented as **bit pattern** (sequence of bits 0s or 1s)
 - Different set of bit pattern used to represent symbols or characters.
 - Each set is called **code**
 - Process of representing symbols is called **encoding**
 - Ex: **ASCII, UNICODE**
- **Numbers**
 - Represented as **bit pattern**
 - Directly converted to **binary form**
- **Audio**
 - Recording or broadcasting of sound or music.
 - Continuous not discrete

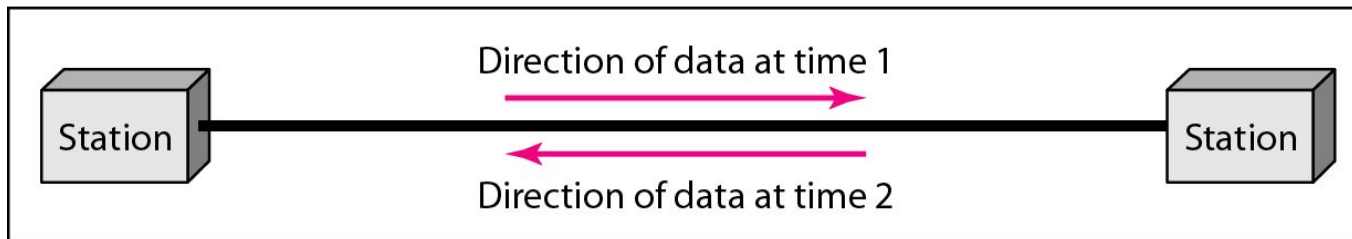
Data Representation (Cont....)

- **Video**
- Recording or broadcasting of picture or a movie
- Produced as :
 - Continuous entity [TV camera]
 - Combination of images-discrete entity
- **Images**
 - Represented as **bit pattern**
 - Image is divided into **matrix of pixels**(smallest element of an image)
 - Each pixel is assigned a **bit pattern** (size and value of pattern depend on image)
 - Ex: black and white dots (chessboard) -1 bit pattern is enough to represent a pixel, gray scale- 2 bit pattern.
 - Several methods to represent colour images : RGB,YCM

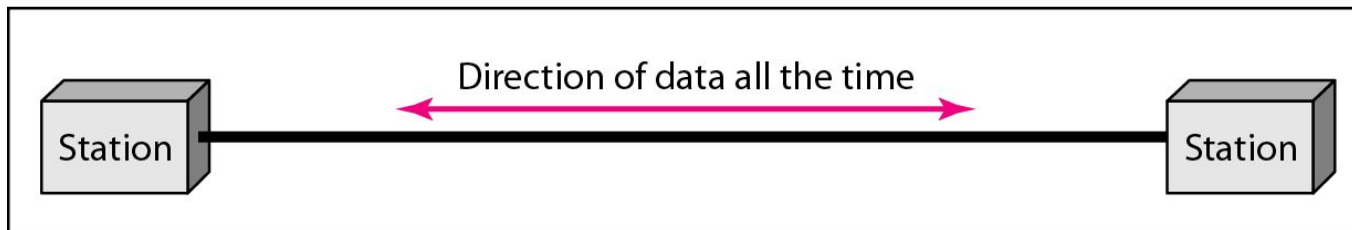
Data flow (simplex, half-duplex, and full-duplex)



a. Simplex



b. Half-duplex



c. Full-duplex

1.2 NETWORKS

- A **network** is a set of devices (often referred to as **nodes/host**) connected by communication **links**.
- A **node** can be a computer, router, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- A **link** can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

Topics discussed in this section:

- Network Criteria
- Physical Structures
- Categories of Networks

Network Criteria

- **Performance**

- **Measured using:**

- **Transit time:** time taken to travel a message from one device to another.
 - **Response time:** time elapsed between inquiry and response.

- **Depends on following factors:**

- Number of users
 - Type of transmission medium
 - Efficiency of software

- **Evaluated by 2 networking metrics:**

- Throughput (high)
 - Delay (small)

Network Criteria (cont..)

- **Reliability**

- **Measured by**

- Frequency of failure.
 - Time taken to recover from a network failure.
 - Network robustness in a disaster.

- **Security**

- **Protecting** data from unauthorized access, damage and development.
 - **Implementing** policies and procedures for recovery from breaches and data losses.
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Physical Structures

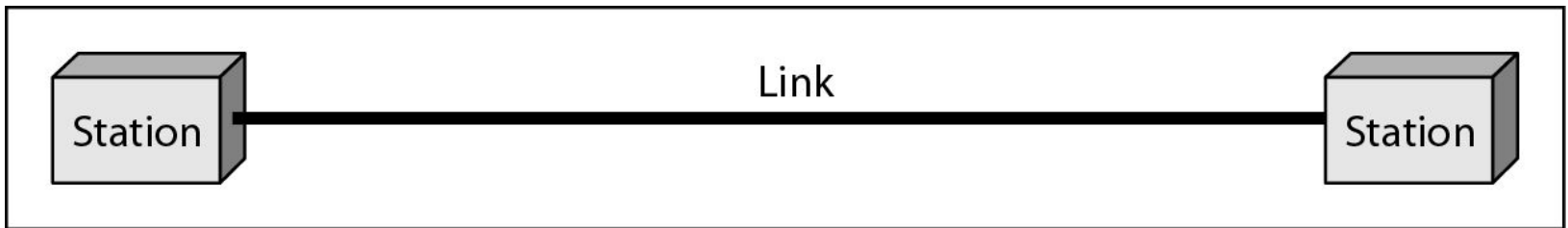
- **Type of Connection**

- **Point to Point**

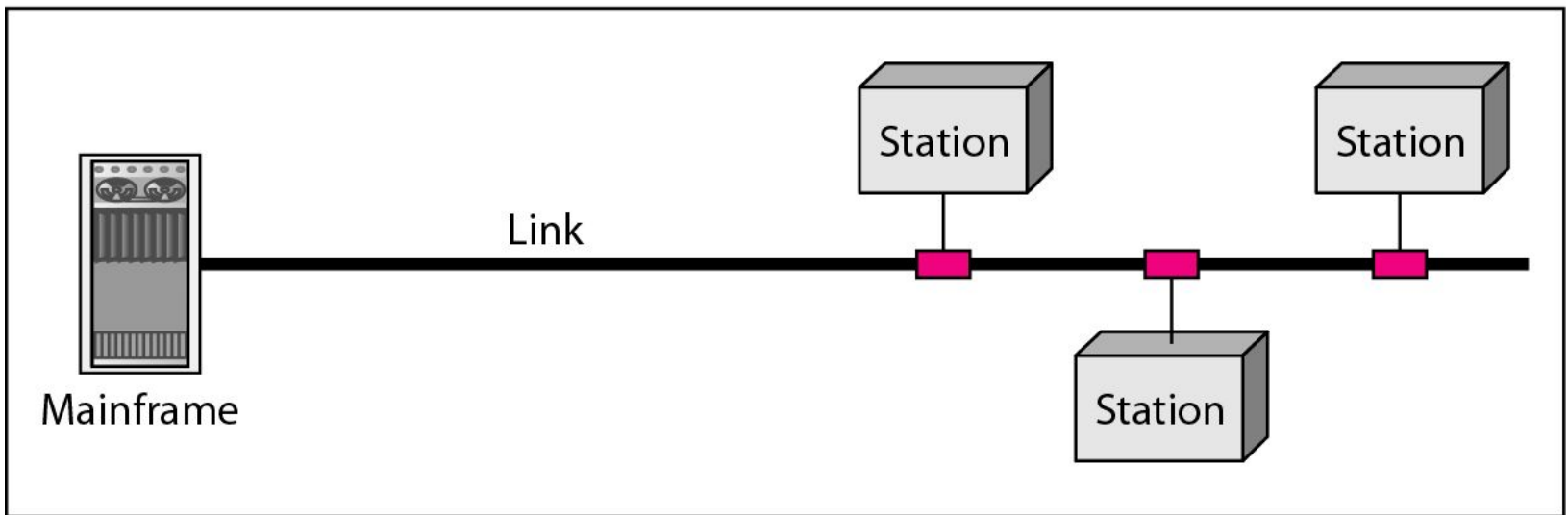
- **single transmitter and receiver**
 - **provides a dedicated link between two devices**
 - **entire capacity of the link is reserved for transmission between those two devices**

- **Multipoint (multidrop)**

- **multiple recipients of single transmission**
 - **more than two specific devices share a single link**
 - **capacity of the channel is shared, either spatially or temporally**
-



a. Point-to-point



b. Multipoint

Quiz

- A _____ connection provides a dedicated link between two devices
 - Point to point
 - Multipoint
 - Both
 - None of them

- Answer: point to point

Quiz

- changing television channels by infrared remote control is an example of
 - Multidrop connection
 - Point to point
 - Both
 - None of them

Answer: Point to point

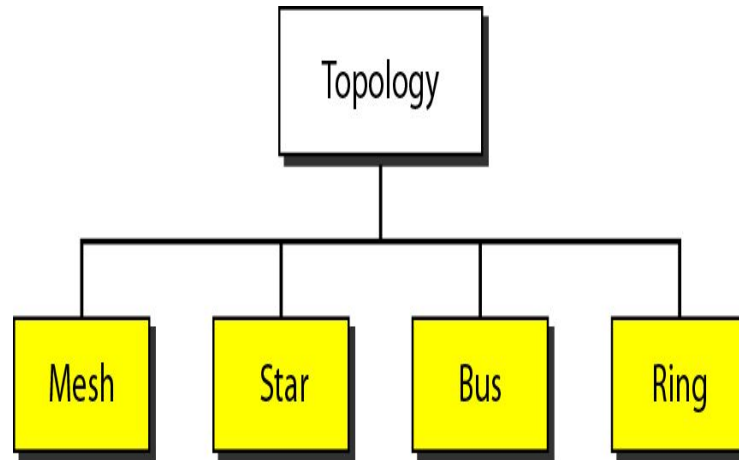
Physical Structures (Cont..)

- **Physical Topology:**

- Way in which network is laid out physically.
- Two or more links form a topology.
- “*Topology of network is the geometric representation of all links and linking devices to one another*”.

- **Basic topologies:**

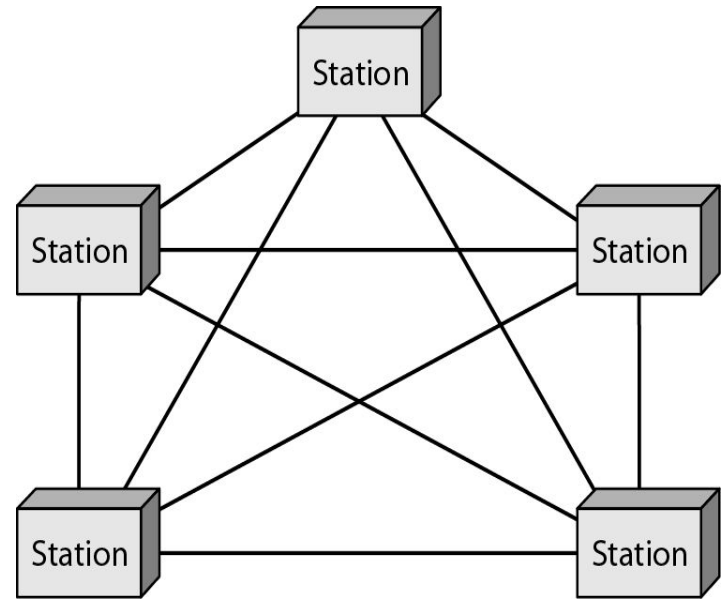
- *Mesh*
- *Star*
- *Bus and*
- *Ring*



Mesh

Topology

- All devices are connected to each other
- Dedicated **point-point link** between all devices
- “ **$n(n-1)$** ” physical channels to link “ **n** ” device.
- For ‘ **n** ’ nodes
 - **$n(n-1)$** physical links
 - **$n(n-1)/2$** duplex mode links
- Every device have $(n-1)$ I/O ports to be connected to other $(n-1)$ devices.



Fully connected Mesh Topology

Mesh Topology (cont..)

Advantages:

- 1) **Congestion reduced:** Each connection can carry its own data load.
- 2) **Robustness:** If one link fails, it does not affect the entire system.
- 3) **Security:** When a data travels on a dedicated-line, only intended-receiver can see the data.
- 4) **Easy fault identification & fault isolation:** Traffic can be re-routed to avoid problematic links.

Disadvantages:

- 1) Difficult **installation and reconfiguration.**
- 2) **Bulk of wiring** occupies more space than available space.
- 3) **Very expensive:** hardware required to connect each link is expensive.

practical example: connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

Quiz

1. Assume that 7 devices are connected in a mesh topology. How many duplex links are needed?

- 42
- 7
- 21

Answer: 21

Quiz

2. Assume that 7 devices are connected in a mesh topology. How many ports are needed?

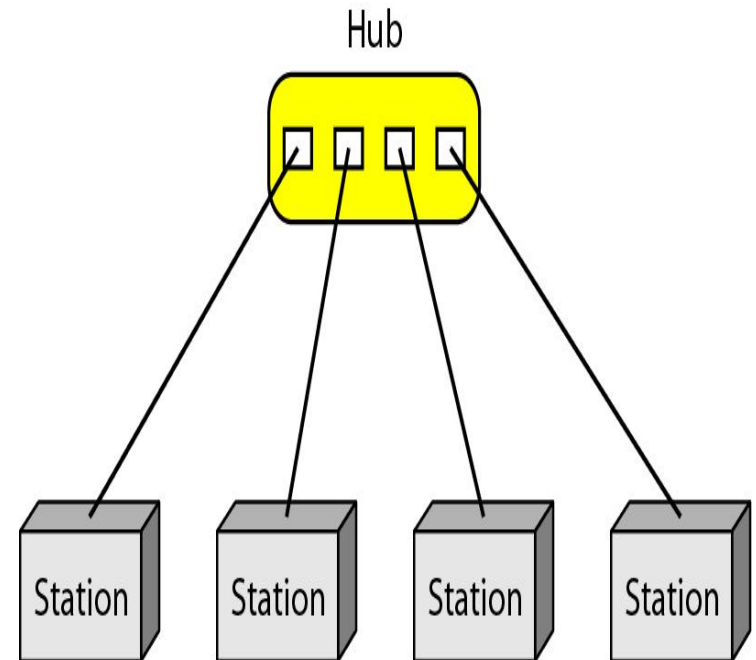
- 5
- 6
- 7

Answer: 6

Star

Topology

- **Point to Point** connection
- All the devices are connected to a central controller called a **hub**
 - Dedicated point-to-point link between a device & a hub.
 - The devices are **not directly linked** to one another. Thus, there is no direct traffic between devices.
- The hub acts as a **junction**:
 - If device-1 wants to send data to device-2,
 - the device-1 sends the data to the hub, then the hub relays the data to the device-2.



A star topology connecting four stations

Star Topology (cont..)

Advantages:

- 1) **Less expensive:** Each device needs only one link & one I/O port to connect it to any devices.
- 2) **Easy installation & reconfiguration:** Nodes can be added/removed w/o affecting the network.
- 3) **Robustness:** If one link fails, it does not affect the entire system.
- 4) Easy to **detect** and **troubleshoot** fault.
- 5) **Centralized management:** The hub manages and controls the whole network.

Disadvantages:

- 1) **Single point of failure:** If the hub goes down, the whole network is dead.
- 2) Cable length required is the more compared to bus/ring topologies.
- 3) **Number of nodes** in network depends on **capacity of hub**.

Example: Local area network

Quiz

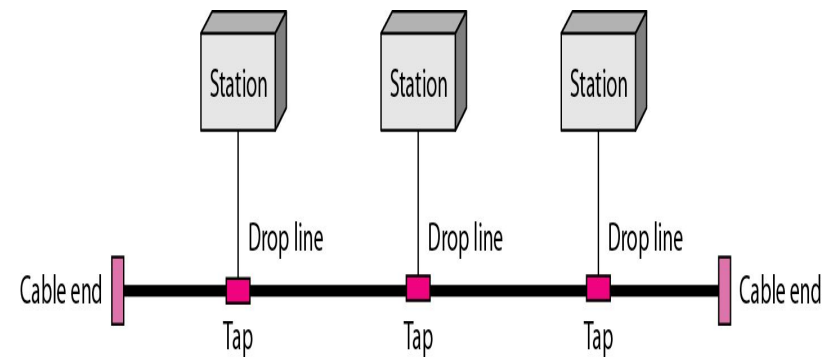
- In a star topology with six computers, we need _____ links.
 - 6
 - 1
 - 5
 - None of these

Answer : 6

Bus

Topology

- **Multipoint connection**
 - All the devices are connected to the single cable called **bus (backbone)**
 - Devices are connected to the bus by drop-lines and taps.
 - A drop-line is a connection running between the device and the bus (main cable).
 - A tap is a connector that links to the bus
 - **Limit on number of taps** a bus can support and **distance** between those taps-(as signal travels along backbone some energy is transferred to heat which makes it weaker as it travels farther)



A bus topology connecting three stations

Bus

Topology

- 1) Easy installation.
- 2) Cable required is the least compared to mesh/star topologies.
- 3) Redundancy is eliminated.
- 4) Costs less (Compared to mesh/star topologies).
- 5) Mostly used in small networks. Good for LAN.

Disadvantages:

- 1) Difficult to detect and troubleshoot fault.
 - 2) Signal reflection at the taps can cause degradation in quality.
 - 3) A fault/break in the cable stops all transmission.
 - 4) There is a limit on
 - i) Cable length
 - ii) Number of nodes that can be connected.
 - 5) Security is very low because all the devices receive the data sent from the source.
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Quiz

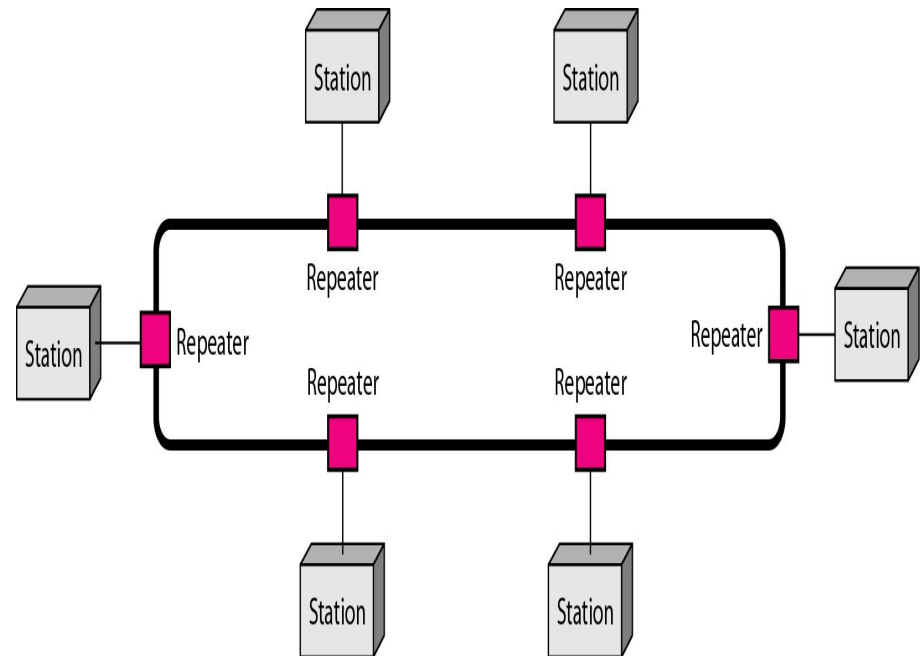
- Bus topology is a
 - Point to point connection
 - Multipoint connection
 - Both
 - None of the above

Answer: Multipoint Communication

Ring

Topology

- Each device is connected to the next, forming a ring
- There are only two neighbours for each device.
- Data travels around the network in one direction till the destination is reached.
- Sending and receiving of data takes place by the help of token.
- Each device has a repeater.
- A repeater
 - receives a signal on transmission-medium &
 - regenerates & passes the signal to next device.



A ring topology connecting six stations

Ring

Topology

1) Easy installation and reconfiguration.

To add/delete a device, requires changing only 2 connections.

2) Fault isolation is simplified.

If one device does not receive a signal within a specified period, it can issue an alarm.

The alarm alerts the network-operator to the problem and its location.

3) Congestion reduced: Because all the traffic flows in only one direction.

Disadvantages:

1) Unidirectional traffic.

2) A fault in the ring/device stops all transmission.

The above 2 drawbacks can be overcome by using dual ring.

3) There is a limit on

i) Cable length &

ii) Number of nodes that can be connected.

4) Slower: Each data must pass through all the devices between source and destination.

Quiz

- Flow of Data in a computer ring topology is
 - Half duplex
 - Duplex
 - Unidirectional
 - Bidirectional

Answer: Unidirectional

Categories of Networks/Network Types

Few criteria –size, geographical coverage and ownership to make this distinction.

- **Local Area Networks (LANs)**
 - Short distances
 - Designed to provide local interconnectivity
 - **Wide Area Networks (WANs)**
 - Long distances
 - Provide connectivity over large areas
 - **Metropolitan Area Networks (MANs)**
 - Provide connectivity over areas such as a city, a campus
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Local Area Networks (LANs)

- It is used to connect computers in a **single office, building or campus**.
- **Privately owned network**.
- A LAN can be simple or complex.
 - 1) **Simple**: LAN may contain 2 PCs and a printer.
 - 2) **Complex**: LAN can extend throughout a company.(extend to audio and video devices.)
- Each host in a LAN has an **address** that uniquely defines the host in the LAN.
- A packet sent by a host to another host carries both **source host's and destination host's addresses**.

Wide Area Networks (WAN)

- WAN can cover **wider geographical area**. It can cover cities, states, countries and even world.
- WAN interconnects connecting devices such as **switches, routers, or modems**.
- Normally, WAN is
 - created & run by communication companies (Ex: BSNL, Airtel)
 - leased by an organization that uses it.

Two distinct examples of WANs:

- **Point-to-point WAN**
- **Switched WAN**

Important Questions

1. Describe data communication and its components.(jan 20, jan 19, jul 19)
2. Assume 5 devices are connected in a mesh topology. How many duplex link is needed? How many ports are needed for each(2M)(jan 18)
3. Explain data flow techniques of data communication(5M)(jan 2020, July 2019)
4. Explain any two fundamental characteristics of communication and basic network topology(8M)(July 2018, July 2019)

5. List out the network criteria(July 2018)
6. What are the factors that determine whether a communication is LAN or WAN?
(July 2019)